# Matteo Ceradini

## AI ENGINEER


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## Summary


I am an **AI engineer** with a **PhD in Biorobotics** and a background as **computer scientist**, specialized in **deep learning and neural signal decoding** (EEG/EMG, BCI). I have **developed machine learning models** for complex signal processing tasks and worked on **RAG systems**, as well as **data and context engineering** across research and industry settings. I am particularly interested in **applying AI to challenging, real-world problems** and in exploring its potential across diverse domains.


## Education

**PhD in Biorobotics**
**Sant'Anna School of Advanced Studies, 2021 - 2025**
Brain–Computer and Body-Machine Interfaces (BCI/BMI); Deep learning for neural signal processing & decoding; Artificial Intelligence and Virtual Reality for human-machine interactions.

**Master in Computer Science (Artificial Intelligence course)**
**University of Torino, 2018-2021**
Grade: 110/110 with honors and distinction.

**Bachelor in Computer Science**
**University of Torino, 2015-2018**
Grade: 103/110.

## Tech stack

| | | |
|---|---|---|
| Python (advanced) | Langchain (medium) | Tensorflow (basic) |
| PyTorch (advanced) | C# (medium) | MongoDB (basic) |
| Github (advanced) | Unity 3D (medium) | NodeJS (basic) |
| Matlab (advanced) | Docker (medium) | ReactJS (basic) |
| SQL (advanced) | | R (basic) |
| Java (advanced) | | C/C++ (basic) |

## Languages

Italian - Native
English - fluent (C1 level)

## Experiences

**AI & Software Engineer**
**ION Group, Dec 2025 - current**
Developed chatbot agents using LangChain, leveraging knowledge graph databases (e.g., FalkorDB) to support bank operators in assessing legal signatory authority.
Contributed as a software engineer to the development of banking solutions for credit management and portfolio analysis.

**Visiting PhD in Biomedical Engineering**
**University of Michigan, Jul 2024 - Feb 2025**
Visiting period in the Chestek Lab, working on the development of online BCI exploiting the usage of implantable neural signals. Main activities inclued ponline decoding of individual fingers movements from implanted EMG signal in non-human participants.

**AI engineer**
**Cynexo & SISSA, May 2021 - Sep 2021**
In collaboration with the Time Perception Lab at SISSA research center I contributed to two main projects. The first involved implementing DeepLabCut for real-time tracking and behavioral tracking of mice in neuroscientific experiments. The second focused on developing a PoC closed-loop system integrating deep-learning models for real-time brain EEG signal decoding targeting the application of controlling transcranial stimulation.

**AI engineer**
**Cynexo & SISSA, Sep 2020 - Mar 2021**
In collaboration with the Visual Neuroscience Lab at SISSA research center I worked on the porting of the DeepLabCut neural network for animal pose estimation to an embedded platform (NVIDIA Jetson). I implemented a knowledge distillation method that resulted in inference speed on the embedded device of ~10fps with an accuracy with an accuracy close to 100% respect to the original model.

## Experiences

**AI Engineer intern**
**Zextras, Jul 2018 - Ago 2018**

Two-month internship at Zextras during the completion of my bachelor's degree. Conducted research on the effectiveness of spam filtering over several thousand emails using deep learning techniques, and implemented a prototype achieving over 98% accuracy with a very low false positive rate.

**Web developer**
**Tecnobit, Jun 2016 - Sep 2020**

From 2016 to 2020, while completing my bachelor's and master's degrees, I worked part-time as a full-stack web developer. My responsibilities included both front-end and back-end tasks, using php frameworks such as Codeigniter and Laravel. During this time, I maintained and developed websites such as sketchupitalia.it, corsigeometri.it, topgeometri.it, and others.

**Summer internships**
**Margraf project & Industrie Metalpress, Jun-Sep 2013 & Jun-Sep 2014**

Summer interships at Margraf Project and Industrie Metalpress (2013-2014). At Margraf, I worked as a marble operator on polishing and finishing tasks. At Metalpress, I worked on company Intraned web-app and on developing a C# application for network security activity monitoring via SNMP protocol.

## Certifications

**Retrieval Augmented Generation (RAG)**
**DeepLearning.ai, Sep 2025**

End-to-end RAG system design and implementation; retrieval strategies, knowledge base integration, and optimization trade-offs (cost, latency, quality)

**Fundamentals of Reinforcement Learning**
**University of Alberta, Alberta Machine Intelligence Institute by Coursera, Oct 2021**

Markov Decision Processes, exploration–exploitation trade-offs, value function methods, and dynamic programming for control and optimization

## Awards & Fellowships

**Best paper presented by a young researcher**
**IEEE MetroXRAINE 2023 Conference**

This award recognizes the most outstanding paper presented by a Young Researcher at IEEE MetroXRAINE 2023.

**Zegna Scholarship Fellowship**
**IEEE MetroXRAINE 2023 Conference**

Recipient of the Zegna Scholarship, funding my 8-months research stay at the University of Michigan where I worked on Brain and Body Machine Interfaces using implantable signals for neuroprosthetics applications.

## Publications on international journals

**The effect of User Learning of Online EEG Decoding of Upper-Limb Movement Intention**
Ceradini et. al. 2025
IEEE Transactions on Medical Robotics and Bionics

**A Virtual Reality-Based Protocol to Determine the Preferred Control Strategy for Hand Neuroprostheses in People With Paralysis**
Losanno*, Ceradini* et. al. 2024 (* equally contributing)
IEEE Transactions on Neural Systems and Rehabilitation Engineering

**Immersive VR for Upper-Extremity Rehabilitation in Patients with Neurological Disorders: a scoping review**
Ceradini et. al. 2024
Journal of NeuroEngineering and Rehabilitation